



CYTOF XT

THE NEXT EVOLUTION



REIMAGINE CYTOMETRY

AND SEE HOW QUICKLY YOUR CLINICAL RESEARCH MOVES AHEAD.

Developing effective therapies rapidly, efficiently and with minimal risk is critical to the advancement of precision medicine. However, the complex and unique nature of every patient's immune system poses challenges to achieving this goal. It is no longer sufficient to rely on the cell profiling approaches of the past.

By quickly and easily providing the highest-resolution proteomic cell profiles of any cytometry platform, mass cytometry provides the most reliable and comprehensive way to capture immune complexity.

Mass cytometry generates an unrivaled, high-parameter snapshot of phenotype and function for every cell. This ability has advanced our understanding of disease and therapeutic benefit with novel insights into the function of checkpoint inhibitors,

the impact of drugs on signaling pathways, tumor microenvironment biology in response to immunotherapy and the efficacy of vaccine candidates, to name only a few.

The proven reproducibility of mass cytometry, run-to-run and site-to-site, combined with the robust nature of its metal-tagged antibodies and unique barcoding technology has led to its adoption in scores of clinical and translational research studies as well as by multiple research consortia.

Now, the technology has advanced even further with automated sample acquisition and internal systems monitoring that enable hands-free, flexible operation.

SIMPLIFY THE DESIGN
AND EXECUTION OF DEEP
CELL-PROFILING STUDIES.

STANDARDIZE SAMPLE
ANALYSIS WITH REPRODUCIBLE
WORKFLOWS AND AUTOMATION.

ACCELERATE NOVEL
THERAPEUTIC DEVELOPMENT
TO IMPROVE HUMAN
HEALTH OUTCOMES.

The next evolution of mass
cytometry is here—CyTOF XT™.
Achieve the unimaginable.



CyTOF[®] XT

FROM INNOVATION TO INSIGHT

Translating discoveries with unparalleled speed, precision and reproducibility.
Explore the new capabilities of mass cytometry and imagine the science of tomorrow.

MASS CYTOMETRY powered by CYTOF

"CyTOF is really helpful because you can **identify all cell subsets** as well as the specific proteins that you are interested in while **avoiding** challenges with **fluorophore spectrum overlap**."

—Hema Kothari, PhD

Research Assistant Professor
University of Virginia



"**Analyzing the immune response** of patients with mild-to-moderate disease, severe disease and critical disease **by mass cytometry** allows us to cover the innate and adaptive **immune responses simultaneously**."

—Benjamin Terrier, MD, PhD

Professor of Medicine, Internal Medicine Department of Cochin Hospital



"Where I believe mass cytometry will have the **highest impact** is during **Phase 1 and Phase 2 vaccine studies** in humans by contributing **invaluable information** to the identification of immunological correlates of protection."

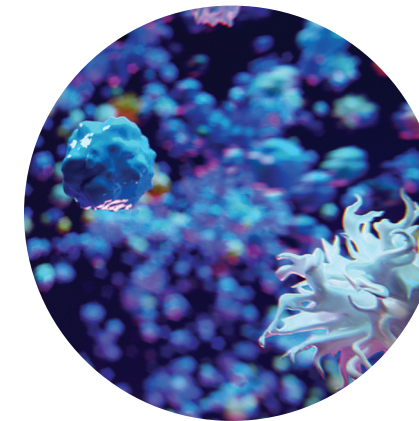
—Marcelo Sztejn, MD, PhD

Associate Director for Basic and Translational Research,
Center for Vaccine Development and Global Health
University of Maryland School of Medicine



XTTRA INSIGHT

ADVANCING UNDERSTANDING OF HUMAN HEALTH AND DISEASE
REQUIRES ASKING THE MOST CHALLENGING QUESTIONS.



Since every second counts in disease diagnosis, the ability to maximize insights into patient biology and successfully execute comprehensive studies could deliver the most value.

The proven reliability, reproducibility and clarity of mass cytometry simplify complex study design and ease site-to-site coordination challenges. By removing these barriers, the technology expedites timelines and accelerates the pace of scientific discovery and study completion. Broad adoption of mass cytometry speaks to the confidence it has inspired for cost-effective clinical study management across sites.

CyTOF XT provides easily scalable solutions that address key hurdles when conducting sophisticated clinical research programs.

SIMPLIFY DEEP CELL
PROFILING.

HARMONIZE SITE-TO-SITE
STUDIES.

ACCELERATE THERAPEUTIC
DEVELOPMENT.

"CyTOF mass cytometry was selected for assessing function and phenotypes of immune-cell subsets ... as it was considered **advantageous over flow cytometry** for the **higher number of biomarkers detected** by antibody panels, with **little or no spillover** between detector channels."

—Chen et al. *Clinical Cancer Research* (2021): "Network for biomarker immunoprofiling for cancer immunotherapy: Cancer Immune Monitoring and Analysis Centers and Cancer Immunologic Data Commons (CIMAC-CIDC)"

CYTOF TECHNOLOGY

CYTOMETRY BY TIME-OF-FLIGHT

Since its commercialization, mass cytometry has revolutionized the cell biology world. Liberated from autofluorescence and fluorescence spillover, mass cytometry simplifies and expands exploration into cellular behavior.

The rapidly evolving environment of clinical and translational research is complex and demanding. Meet these challenges head-on using CyTOF XT to accelerate the path to answers through better agility, customization and execution.

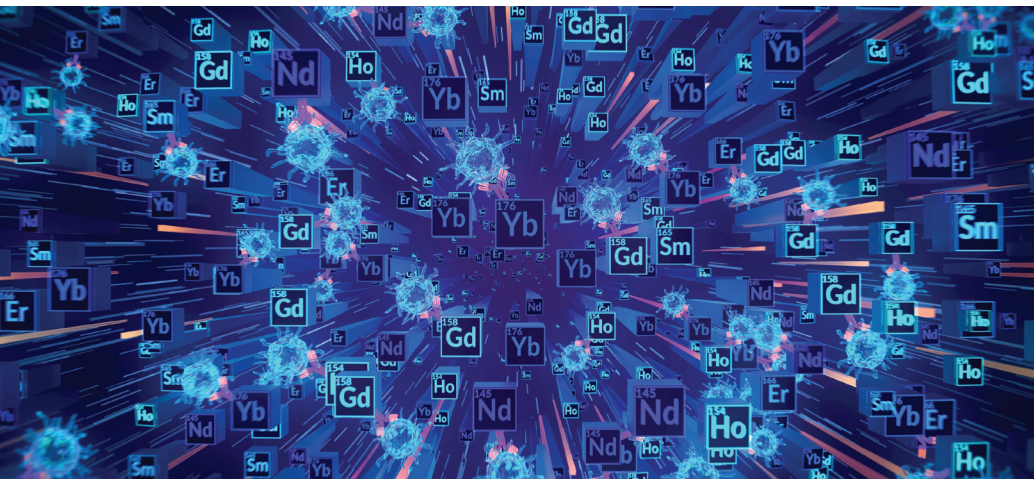
CyTOF XT enables simplified and superior cell profiling, empowering practical biomarker-driven clinical research solutions that can optimize and personalize disease management.

With hands-free sample acquisition, easier operational workflows and reduced cost of installation and ownership, CyTOF XT leads as an ideal platform for a variety of research applications.

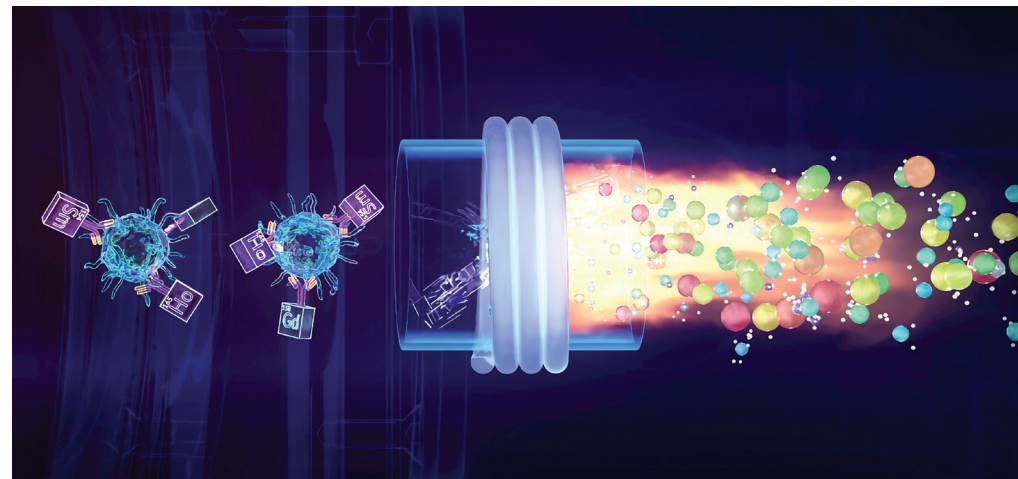
By incorporating mass spectrometry, CyTOF technology powers mass cytometry to quickly provide a high-resolution proteomic profile of each cell, giving a more comprehensive functional and phenotypic view of complex systems from a single tube of precious patient sample.

CYTOF XT

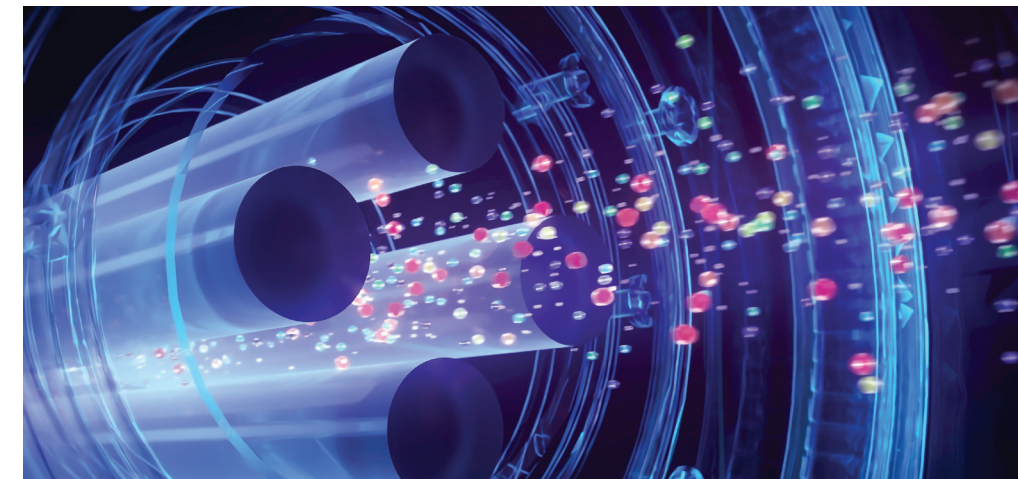
THE SPARK THAT DRIVES INNOVATION



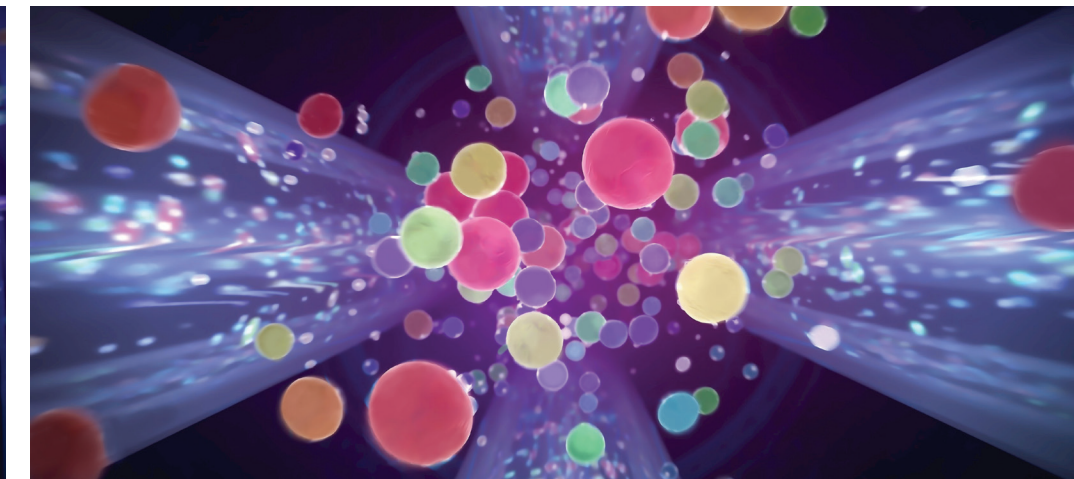
Mass cytometry uses stable metal isotope tags to label cellular targets for precise detection in 135 distinct channels.



More than 50 isotopes can be simultaneously measured per tube of cell sample.



CyTOF technology overcomes the limitations of fluorescence-based detection modalities by separating signals based on differences in mass instead of wavelength.



This powerful single-detector system with straightforward signal path from ionization to detection results in negligible background and minimal signal overlap.

— THINK YOU KNOW CYTOF? THINK AGAIN.

Mass cytometry brings a level of **speed**, **precision** and **reproducibility** to cell analysis not enabled by other single-cell platforms.

CyTOF XT further enhances the capabilities of mass cytometry with workflow automation that refines sample processing and data acquisition.

The instrument offers a new level of autonomy through streamlined operation, automatic monitoring and easier system management, altogether resulting in superior data quality. So you can focus on what matters most.

Adoption of mass cytometry for use in clinical and translational research is rising steadily, year on year.

MULTIPLE RESEARCH CONSORTIA

OVER 60% OF US COMPREHENSIVE CANCER CENTERS

OVER 150 CLINICAL TRIALS USING MASS CYTOMETRY

OVER 1,300 PEER-REVIEWED PUBLICATIONS



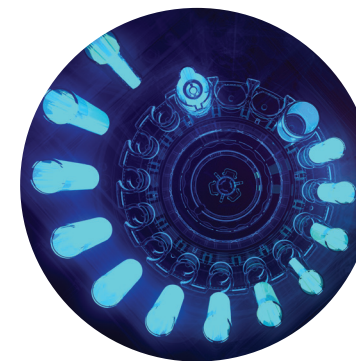
Ignite your research. CyTOF XT

With a simpler approach and cutting-edge technology, **move beyond ordinary to XTraordinary.**

SIMPLIFY. STANDARDIZE. ACCELERATE.

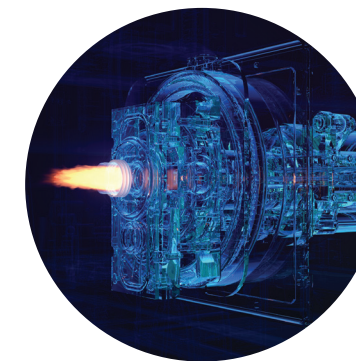
CyTOF XT is the most technologically advanced mass cytometer to date, designed to build on the benefits of CyTOF technology while simplifying operation and automating sample acquisition for the best reproducibility and time to results from every run.

The power of mass cytometry to perform 50-plus-marker high-parameter analysis on a single tube of sample has created a new standard of expectations—that more can always be discovered if you have the right tools.



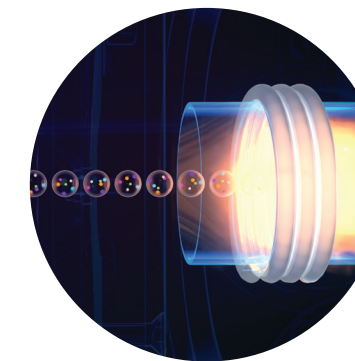
XTra Autonomy

Walk away from sample loading, acquisition and monitoring. Do more with your time.



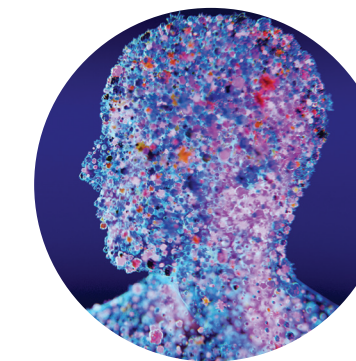
XTra Efficiency

With easy setup and automated system optimization, enjoy peak performance from minimal input.



XTra Clarity

Achieve unrivaled quality and depth of understanding from a greater selection of markers.



XTra Insight

Simpler cell profiling with standardized workflows to accelerate therapeutic development.

XTRA AUTONOMY

Clinical and translational researchers have embraced the power of CyTOF technology to uncover biological insights. CyTOF XT helps you surpass current processes by streamlining and accelerating your approach to efficiently conducting clinical research studies.

CHILLED SAMPLE CAROUSEL

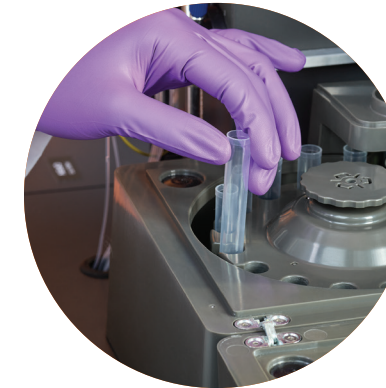
WALK-AWAY ACQUISITION

AUTOMATED CLOG SENSING

WALK-AWAY AUTOMATION

Innovations in CyTOF XT automation expand the possibilities of what your lab can accomplish. The chilled carousel of the Autosampler preserves cell integrity while samples are held for their run. The system automatically resuspends pelleted samples and adds internal standardization beads for improved data quality.

Automated, walk-away sample acquisition can greatly improve reproducibility by minimizing manual error. The automated clog sensing feature of CyTOF XT provides peace of mind and confidence in results, knowing the instrument self-monitors potential issues and addresses them before they occur.



SIMPLE SAMPLE PREP

Consistency of data is key, whether working to prove a hypothesis or standardize a clinical study protocol. The automation and system optimization of CyTOF XT deliver consistent, high-quality data for multiple samples at a time to help fast-track extensive studies.

WHAT IF ... you could **calibrate once** for numerous different samples and panels, and **run hands-free for hours**?



XTRA EFFICIENCY

FASTER BY DESIGN, CYTOF XT SIGNIFICANTLY REDUCES SETUP TIME FOR MORE EFFICIENT SAMPLE PROCESSING AND SHORTER TIME TO RESULTS.



The simplified front-end assembly, bottle tray for solutions and automated tuning minimize maintenance and calibration. Instrument optimization with EQ™ Calibration Beads requires just one reagent to calibrate and optimize the system for any panel.

With the benefit of hands-free operation, mass cytometry with CyTOF XT provides unparalleled efficiency for quality and timely outcomes by incorporating sample sizes of thousands to millions of cells and walkup sample addition for added flexibility.

IMAGINE IF ... your **high-parameter** cytometer needed only **one daily optimization**, applicable to every experiment and **maintained automatically**.

EASY DAILY SETUP

AUTOMATED
SYSTEM TUNING

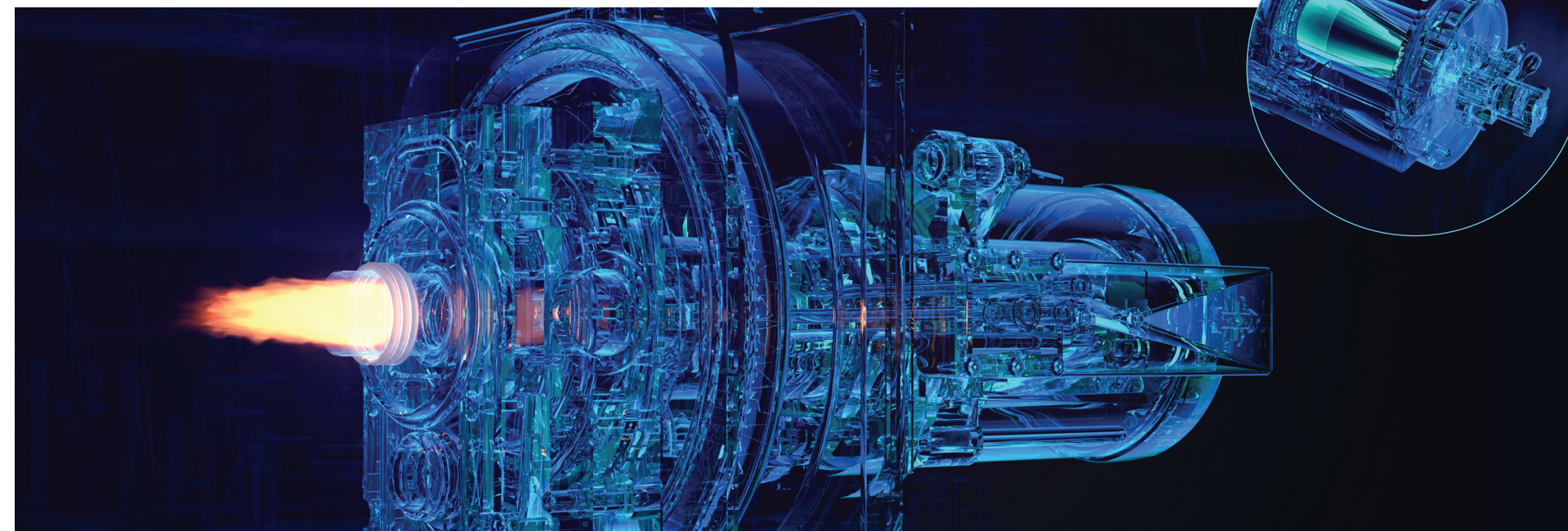
REAL-TIME SIGNAL
OPTIMIZATION

SIMPLE
SETUP

WHAT IF ... your high-parameter data is ready for **analysis immediately after acquisition**?

OPTIMIZE YOUR RESEARCH

CyTOF XT real-time signal optimization stabilizes the detection system for long acquisitions, especially important for multiplexed samples. Variability from acquisition is minimized, and data is processed to FCS files and is ready for analysis right after acquisition with no special data transformations needed.



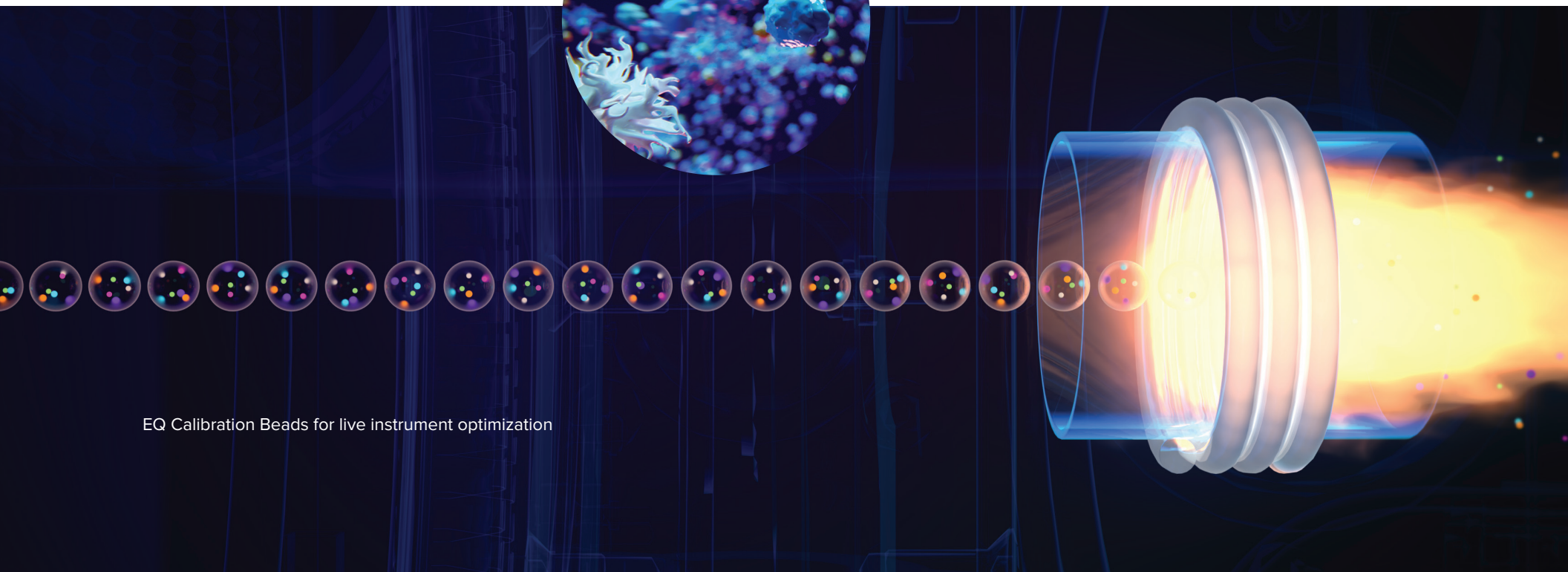
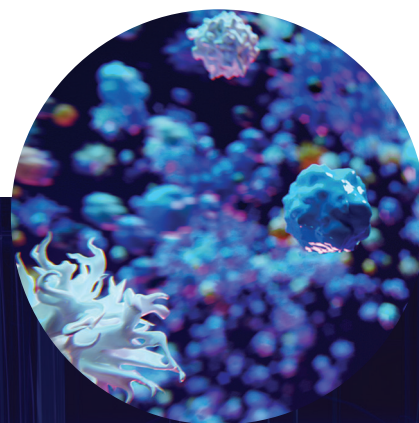
XTRA CLARITY

The precision of measuring marker expression using mass spectrometry is unmatched by fluorescence cytometry methods. Consequently, more accurate data is collected from cleaner signals.

UNRIVALED DATA QUALITY

MOST CYTOMETRIC MARKERS

SUPERIOR CELL PROFILING



EQ Calibration Beads for live instrument optimization

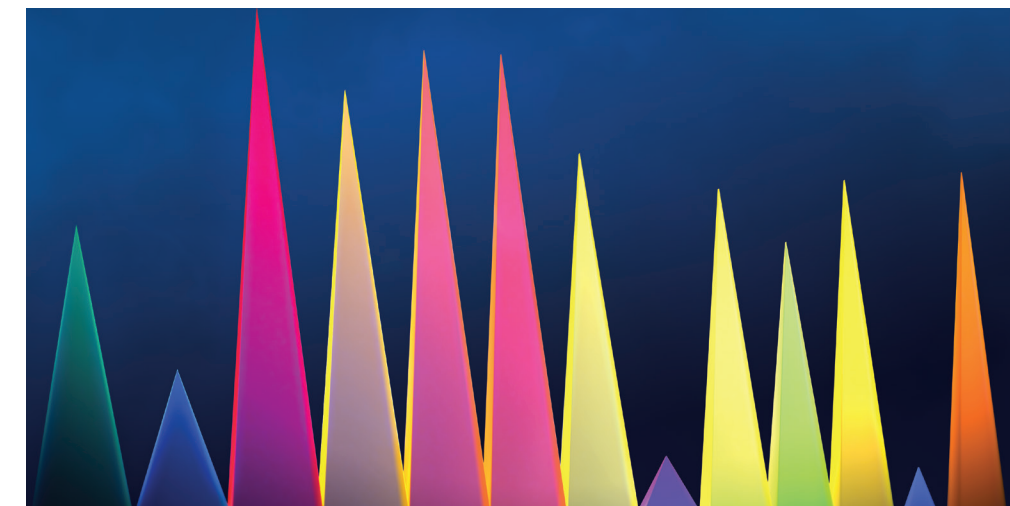
BETTER DATA QUALITY

WHAT IF ... you could design and validate a high-parameter panel **in days, not weeks?**

And then **easily add markers** to modify that panel without heading back to the drawing board?

In mass cytometry, there are no complicated algorithms to apply to deconvolute or compensate for spectral overlap. This equates to less chance of error and data misinterpretation, no need for extra control tubes of sample and far easier 50-plus-marker panel creation and modification.

As a result, mass cytometry is being applied to the broadest range of applications, spanning analysis not only of phenotypic markers but signaling and transcriptional protein expression, deep profiling of T and B cell repertoires including antigen-specificity, and epigenetic changes.



Distinct signal detection results in cleaner data with no need for additional manipulation.

THE MAXPAR DIRECT IMMUNE PROFILING ASSAY



IMMUNE CELL PROFILING WITHOUT COMPROMISE

Gain even more by using the Maxpar® Direct™ Immune Profiling Assay™ on CyTOF XT. The combination of a single-tube workflow with automated multiple sample processing enables a greater level of efficiency, unmatched in conventional protocols.



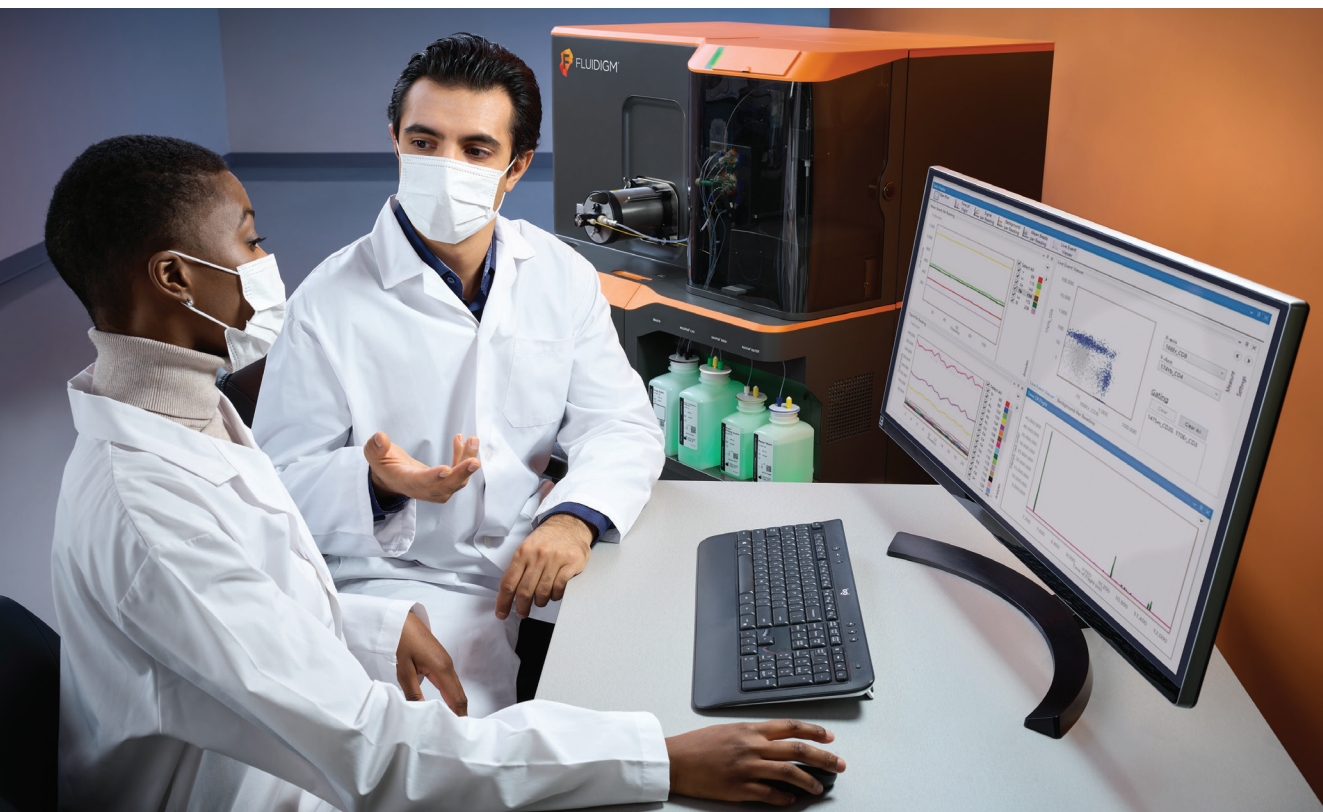
The standardized assay can be harmonized across sites for high-dimensional immune profiling of human PBMC and whole blood, providing the broadest single-tube view of the immune system from each precious sample. Run the assay seamlessly across numerous samples unattended by relying on CyTOF XT for automated sample processing and acquisition.

Extend your view and the possibilities with an easily expandable marker base panel and 5-minute data analysis. Simply count 37 immune cell populations with performance you can trust.

30 MARKERS.

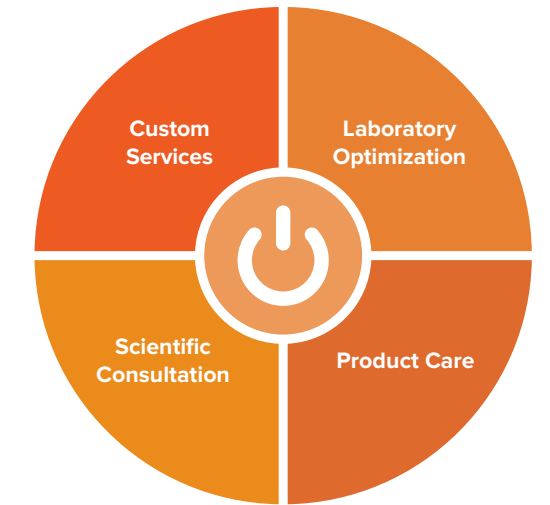
5 MINUTES.

1 TUBE.



WE'VE GOT YOU COVERED WITH **FLUIDIGM PRO SERVICES**

A critical success factor when adopting high-performance technology is acquiring the proficiency and support to accelerate your research and discovery initiatives. Fluidigm PRO Services are a comprehensive portfolio of solutions to help you maximize your investment, scale up productivity and empower your laboratory with greater access to professional resources.



Laboratory Optimization

Enhance operational efficiency and productivity.

Customized services from our diverse team of experts help you prepare for technology adoption, streamline workflow and support continuous improvement initiatives.

Product Care

Maximize your investment and system performance.

Our global network of trained experts provides timely support, training programs, compliance services, proactive maintenance and repair services to optimize productivity and performance of your systems.

Scientific Consultation

Accelerate your research with expert applications support.

Consultative services include experimental and panel design, workflow setup, application troubleshooting and optimization and data analysis.

Custom Services

Leverage tailored services to achieve unique research goals.

Our expert Therapeutic Insights Services team will customize, manage and execute your project from sample to answer, helping to deliver insights from high-dimensional single-cell data.

Providing expertise to power insights

JOIN THE GROWING MASS CYTOMETRY COMMUNITY

Mass cytometry offers an ideal, affordable solution for the routine characterization of target cell populations, with easier panel design and standardized instrument setup to deliver trusted results. CyTOF XT adds an additional layer of efficiency, simplicity and autonomy to truly clarify our understanding of biological systems and clinical outcomes.

Whether your goal is to gain greater knowledge of disease progression, discover more predictive or prognostic biomarker signatures or simplify the path of therapeutic development from bench to bedside, Fluidigm is committed to enabling you to improve the human condition. Join the growing mass cytometry community to further your research and imagine what CyTOF XT can do for you.

Ask for a visit from your local representative to find out more about how CyTOF XT can help you achieve your research goals.

Visit fluidigm.com to see current publications, read interviews with our users, watch videos and learn more about mass cytometry.

WHAT IF ... you could employ **affordable, cutting-edge technology**, accelerating your research **beyond what's currently possible?**

YOU CAN. The new **CyTOF XT**—the next evolution in mass cytometry for translational and clinical research is here. Contact us: fluidigm.com/cytof.xt



CyTOF[®] XT

WHAT INSIGHTS
AWAIT
YOU NEXT?

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